



Document Object Model (DOM) 40m

We'll start by looking at how browsers read HTML, which takes us to the Document Object Model (DOM).

What is the DOM?

According to the WHATWG (2025):

DOM defines a platform-neutral model for events, aborting activities, and node trees.

The DOM is the agreed way for any browser to construct a page from HTML, CSS and JavaScript. Browsers will take the HTML tags and parse them to build a picture of the page, as shown in Example 5.6.



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HTML

Result

EDIT ON

LIVE

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width,
initial-scale=1.0">
  <title>Internetting Is Easy!
</title>
</head>
<body>

  <h1>Internetting Is Easy!</h1>
  <p>First, we need to learn some
basic HTML.</p>

  <h2>Headings</h2>
  <p>Headings define the outline of
your site. There are six levels of
headings.</p>

  <h2>Lists</h2>
  <p>This is how you make an
unordered list:</p>
  <ul>
    <li>Add a <code>&lt;ul&gt;
</code> element (it stands for
unordered list)</li>
    <li>Add each item in its own
  
```

Internetting Is Easy!

First, we need to learn some basic HTML.

Headings

Headings define the outline of your site. There are six levels of headings.

Lists

This is how you make an unordered list:

- Add a `<ul>` element (it stands for unordered list)
- Add each item in its own `<li>` element
- They don't need to be in any particular order

This is what an ordered list looks like:

1. Notice the new `<ol>` element wrapping everything
2. But, the list item elements are the same
3. Also note how the numbers increment on

Resources
1x 0.5x 0.25x
Rerun

Example 5.6 HTML page hierarchy



In the DOM, the HTML tags are objects, and the nested tags (the tags that are organised as sub-tags within a main tag) are children of the main tag. The DOM is often referred to as a DOM tree because it typically consists of a tree-like structure of objects called nodes. A tree is a good metaphor for the DOM because it is easy to understand and visualise the HTML tags being arranged in a tree-like structure.

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A browser will build a DOM tree, which will include several types of objects, such as:

- a document type node
- element nodes
- text nodes
- comment nodes.

Take, for example, the code presented in Example 5.1. This code can be arranged in a DOM tree showing the structure of nodes. Example 5.7 shows the DOM tree for Example 5.1. Select the CSS tab to view the DOM tree.



CSS

Result

EDIT ON

LIVE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width,
initial-scale=1.0">
  <title>Tree Structure</title>
  <style>
    .tree {
      font-family: Arial,
sans-serif;
      white-space: pre;
      max-height: 300px; /*
Set the maximum height */
      overflow-y: auto; /*
Enable vertical scrolling */
      border: 1px solid #ccc;
/* Optional border for clarity */
      padding: 10px; /*
Optional padding */
    }
    .tree ul {
      list-style-type: none;

```

Resources1x0.5x0.25xRerun

Example 5.7 DOM tree

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Can you see the tree-like structure in Example 5.7? You can see that doctype is the root node which has one child node (the HTML element), which then has two child nodes (head and body) and each of these objects then has child nodes of their own.

How does a browser render a web page from HTML?

Rendering is the process where code is turned into a web page that is displayed to users. A rendering engine (software integrated into the browser) completes this process by using the DOM to determine the layout of each element and 'paint' each element on the screen.

Each browser uses a different rendering engine to render the same DOM, but they all follow a similar process.

Watch *How web browsers work* (Petra, 2022) (6:09). The video provides an in-depth explanation of the inner workings of web browsers. It covers the architecture of browsers, including the processes of DNS resolution, DOM tree construction and rendering.



ENTER FULLSCREEN

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How Web Browsers Work

Loading...

Search...



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Now complete the following tasks.
Skip to main content



☒ What is a browser? (1)

Non-Assessed

Status: Done

Reset ↺

1. Read each statement carefully.
2. Mark 'True' if the statement is correct.
3. Mark 'False' if the statement is incorrect.

	True	False
The main function of a browser is to present web resources by requesting them from the server.	☒	☐
HTML can be parsed easily by conventional parsers because its grammar is context-free.	☐	☒

1. The main function of a browser is to present web resources by requesting them from the server.
 - True, browsers request and display web resources.
2. HTML can be parsed easily by conventional parsers because its grammar is context-free.
 - False, HTML's forgiving nature makes it difficult for conventional parsers.

Previous attempts ▼

[Skip to main content](#)



☒ What is a browser? (2)

Non-Assessed

Status: Done

Reset ↺

1. Read each statement carefully.
2. Mark 'True' if the statement is correct.
3. Mark 'False' if the statement is incorrect.

	True	False
The DOM tree is constructed of elements that implement one of the DOM interfaces.	☒	☐
CSS parsing cannot be parsed using regular parsers	☐	☒
The rendering engine is responsible for displaying requested content on the browser screen	☒	☐

1. The rendering engine is responsible for displaying requested content on the browser screen.
 - True, the rendering engine handles content display.
2. CSS parsing cannot be parsed using regular parsers.
 - False, CSS uses context-free grammar for parsing.
3. The DOM tree is constructed of elements that implement one of the DOM interfaces.
 - True, DOM elements implement specific interfaces.

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Previous attempts ▼

☒ What is a browser? (3)

Non-Assessed

Status: Done

Reset ↺

1. Read each statement carefully.
2. Mark 'True' if the statement is correct.
3. Mark 'False' if the statement is incorrect.

	True	False
The layout process involves calculating the position and size of each renderer	☒ ✓	☐
Speculative parsing is an optimisation technique used by browsers to load resources in parallel	☒ ✓	☐
The painting process consists of directly updating the DOM tree and bypassing the render tree, without invoking the paint() method on any renderer.	☐	☒ ✓

1. Speculative parsing is an optimisation technique used by browsers to load resources in parallel.
 - True, speculative parsing improves loading efficiency.

2. The layout process involves calculating the position and size of each renderer.

Skip to main content ☒ True, layout calculates positions and sizes.

3. The painting process consists of directly updating the DOM tree and bypassing the render tree, without invoking the `paint()` method on any renderer.

- False, during the painting step, the system goes through the render tree to draw each element on the screen.

Previous attempts ▼



☒ What is a browser? (4)

Non-Assessed

Status: Done

Reset ↺

1. Read each statement carefully.
2. Mark 'True' if the statement is correct.
3. Mark 'False' if the statement is incorrect.

	True	False
The CSS box model describes the rectangular boxes generated for elements in the document tree	☒	☐
The event loop in a browser's main thread runs only once and does not process any events	☐	☒

1. The event loop in a browser's main thread runs only once and does not process any events.

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- The event loop continuously processes events.

2. The CSS box model describes the rectangular boxes generated for elements in the document tree.

- True
- The CSS box model defines element boxes.

Previous attempts ▼



The next section will explore HTML in more depth.



Thoughts on this page?

